Heart Disease Prediction Using Machine Learning

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Introduction

What is heart disease?

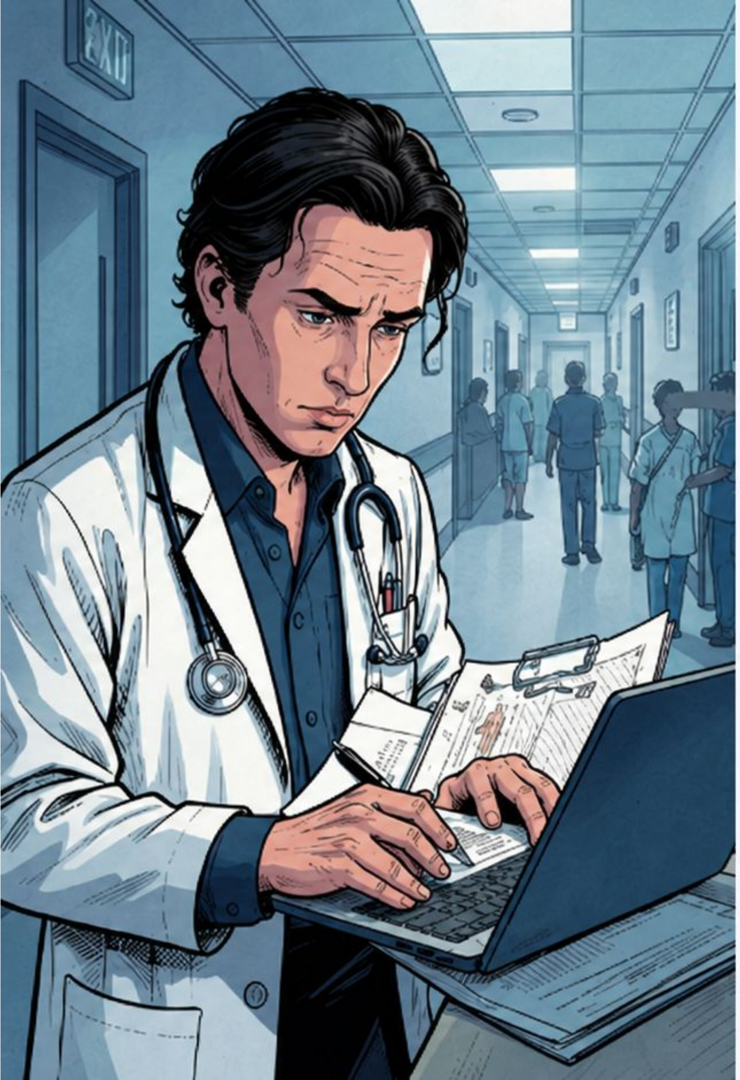
Collective term for conditions affecting the heart – coronary artery disease, arrhythmia, heart failure – often silent until advanced.

Importance of early prediction

Early detection reduces morbidity and enables timely interventions that save lives and lower treatment costs.

AI & ML in healthcare

Machine learning analyses patterns in medical data to assist clinicians with risk stratification and decision support.



Problem Statement

Heart disease is a leading cause of death worldwide. Early diagnosis is challenging due to subtle symptoms and variable presentations. There is a pressing need for an intelligent, data-driven prediction system to flag high-risk patients for further evaluation.

Objectives

Data analysis

Explore and understand patterns within the heart disease dataset.

Preprocessing

Clean, impute, normalise and encode features for modelling.

Model building

Train Logistic Regression and Random Forest classifiers.

Evaluation & comparison

Compare accuracy and confusion matrices; select the final model.

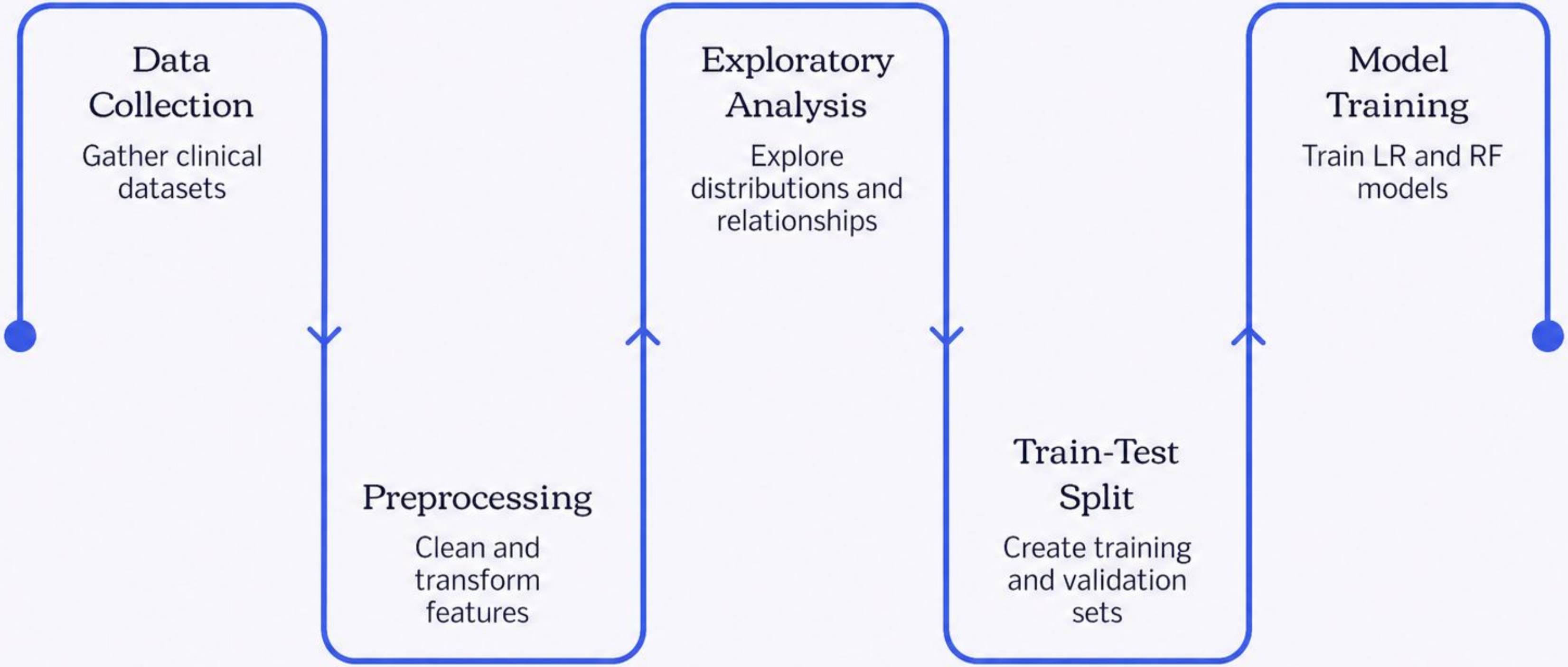
Prediction system

Produce a deployable pipeline to predict heart disease risk.

Dataset Description

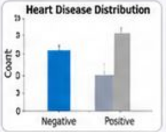
Property	Details
Total records	303 patient records
Total features	14 attributes (clinical and demographic)
Key attributes	Age, Sex, Chest Pain Type, Resting Blood Pressure, Serum Cholesterol, Fasting Blood Sugar, Resting ECG, Maximum Heart Rate, Exercise-induced angina, ST depression, Slope, Vessels, Thalassemia, Target
Target	Binary label indicating presence (1) or absence (0) of heart disease
Usage	Supervised classification – predict target from clinical features

Methodology



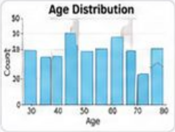
End-to-end pipeline: collect clinical data, clean and transform features, explore distributions and relationships, split data, train models (Logistic Regression, Random Forest), evaluate performance and prepare the prediction pipeline for deployment.

Data Visualisation – Key Plots



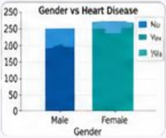
Heart Disease Distribution

This plot shows the proportion of positive and negative cases in the dataset. The split appears balanced to moderately imbalanced, which is important context when selecting evaluation metrics and interpreting model performance.



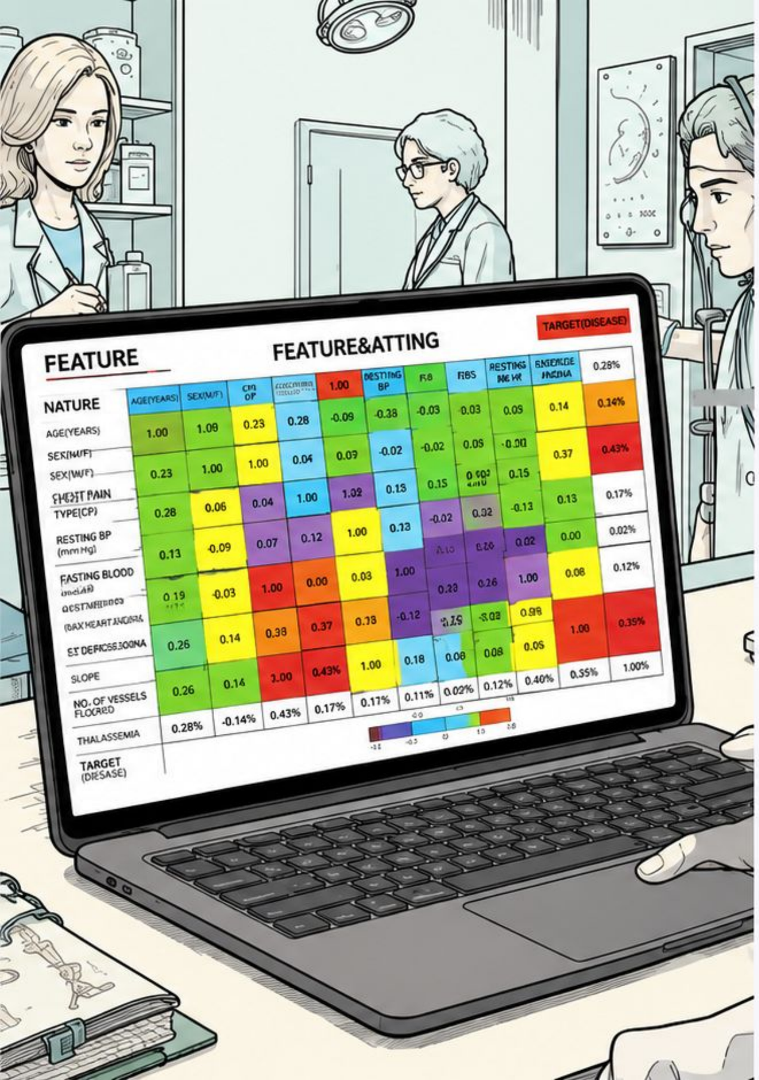
Age Distribution

Age is concentrated around middle-aged and older adults, suggesting the sample reflects a population where cardiovascular risk is more common. This pattern helps frame the analysis and supports expectations about how risk factors may vary across age groups.



Gender vs Heart Disease

This comparison highlights differences in heart disease prevalence across genders within the sample. It is useful for stratified analysis and may point to meaningful feature interactions worth examining further.



Correlation Analysis

Heatmap reveals relationships between features. Examples:

- Positive correlations: Age ↔ Max heart rate decline (age-related risk patterns).
- Negative correlations: HDL-related patterns vs risk markers (illustrative).
- Target correlation: Attributes like chest pain type and ST depression often show stronger correlation with the target.

Understanding these relationships helped with feature selection, multicollinearity checks and improving model generalisability.

Results & Performance

86.89%

Logistic Regression

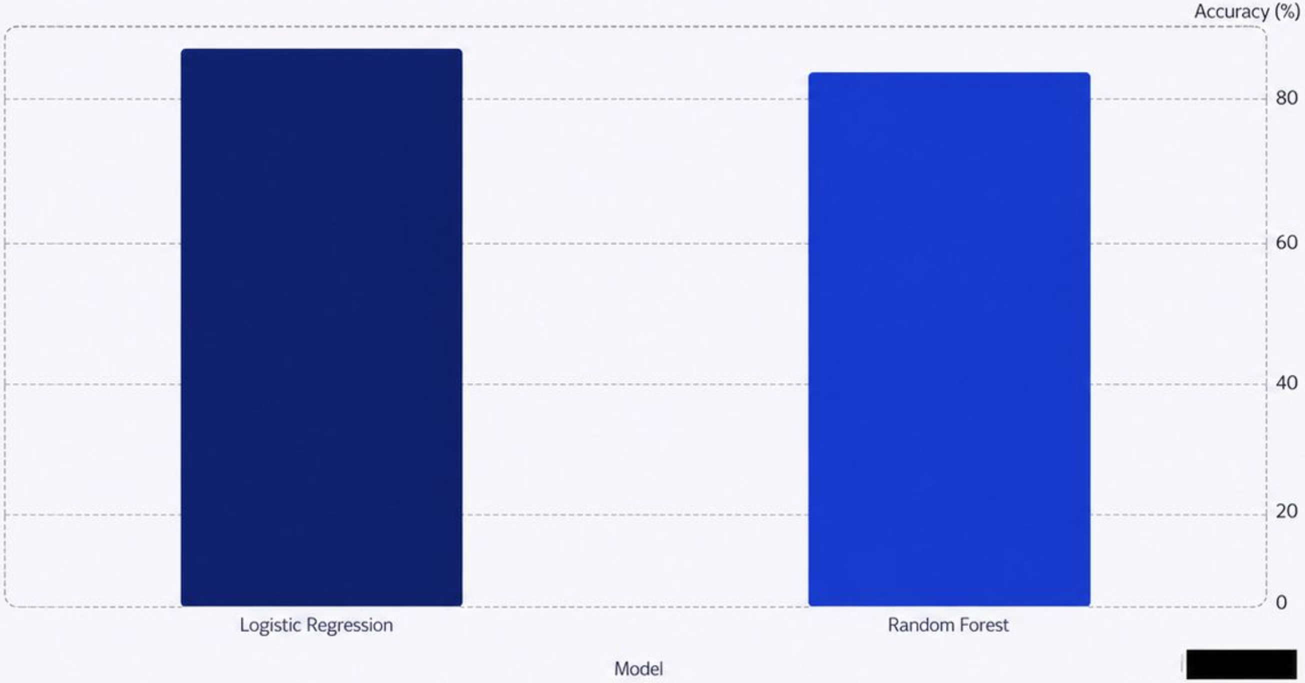
Final evaluation model with overall accuracy on test set.

83.61%

Random Forest

Strong performance, slightly lower accuracy than Logistic Regression in this dataset.

Confusion matrix exploration: True Positives correct disease detection; True Negatives correct non-disease; False Positives flags normal as risk; Negative class F1 score balances false negatives for efficient alerts.



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Model comparison chart above – Logistic Regression selected due to superior accuracy and simpler interpretability for clinical settings.

Conclusion & Future Scope

1

Conclusion

Developed a reliable heart disease prediction system from a 303-record dataset. Logistic Regression achieved 86.89% accuracy and was chosen for final deployment due to performance and interpretability.

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Future Scope

Planned enhancements include integration with hospital systems, a mobile app for clinician use, real-time patient monitoring, expanded datasets and continuous learning for improved accuracy.

For deployment considerations: ensure clinical validation, data privacy, and explainability before integration into care pathways.